

(TR-104) PYTHON WITH AI/ML

Training Day 27 Report

Date: 17 February 2026

Topic Covered: Advanced NLP Preprocessing, TF-IDF Tuning & Job Recommendation Engine

The twenty-seventh day of training focused on improving NLP pipelines using advanced preprocessing techniques, TF-IDF tuning, similarity-based retrieval systems, and job recommendation logic using cosine similarity.

- Built an improved Resume / Job Description Classifier using advanced NLP preprocessing techniques.
- Performed text cleaning by removing emails, numbers, punctuation, and extra spaces from resume text.
- Applied tokenization and stopwords removal for efficient text preprocessing.
- Implemented lemmatization to convert words into meaningful base forms.
- Compared unigram and bigram models using TF-IDF vectorization.
- Tuned TF-IDF parameters including max_features, min_df, and max_df for better performance.
- Performed proper train-test data splitting to avoid data leakage.
- Trained and evaluated Logistic Regression model using processed resume data.
- Evaluated model performance using F1-score and other NLP evaluation metrics.
- Studied cosine similarity and similarity score interpretation in NLP systems.
- Built a Job Recommendation Engine that recommends top matching job roles based on resume text similarity.
- Implemented ranking system for displaying top matching jobs with similarity scores.
- Developed reusable similarity search function for matching resumes with job descriptions.
- Handled edge cases such as short resumes, unrelated resumes, and mixed-domain resumes.

- Strengthened understanding of search engine style retrieval and semantic matching concepts.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/Resume-Job-Classifier>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/from-text-to-talent-a-resume-classification-system-using-nlp-89f441ca9902>